the likelihood that at least one of them impacts the Earth is probably increased, and the apocalyptic character of the consequences may not be much reduced. On the other hand, if the explosion disrupts the asteroid into a swarm of objects a hundred meters in diameter or smaller, all of them might ablate away like giant meteors on entering the Earth's atmosphere. In this case little impact damage would be caused. Even if the asteroid were wholly pulverized into fine powder, though, the resulting high-altitude dust layer might be so opaque as to block the sunlight and change the climate. We do not yet know.

A vision of dozens or hundreds of nuclear-armed missiles on ready standby to deal with threatening asteroids or comets has been offered. However premature in this particular application, it seems very familiar; only the enemy has been changed. It also seems very dangerous.

The problem, Steven Ostro of JPL and I have suggested, is that if you can reliably deflect a threatening worldlet so it does not collide with the Earth, you can also reliably deflect a harmless worldlet so it *does* collide with the Earth. Suppose you had a full inventory, with orbits, of the estimated 300,000 near-Earth asteroids larger than 100 meters—each of them large enough, on impacting the Earth, to have serious consequences. Then, it turns out, you also have a list of huge numbers of inoffensive asteroids whose orbits could be altered with nuclear warheads so they quickly collide with the Earth.

Suppose we restrict our attention to the 2,000 or so near-Earth asteroids that are a kilometer across or bigger—that is, the ones most likely to cause a global catastrophe. Today, with only about 100 of these objects catalogued, it would take about a century to catch one when it's easily deflectable to Earth and alter its orbit. We think we've found one, an as-yet-unnamed* asteroid so far denoted only as 1991OA. In 2070, this world, about 1 kilometer in diameter, will come within 4.5 million kilometers of the Earth's orbit—only fifteen times the distance to the Moon. To deflect 1991OA so it hits the Earth, only about 60 megatons of TNT equivalent needs to be exploded in the right way—the equivalent of a small number of currently available nuclear warheads.